TEST REPORT DOCUMENT

Project Name	*Develop an AI-Assisted Omnichannel Order Fulf*	**Creator**	Nguyễn Phương Mỹ Thuận
Project Code	*FA26SE036*	**Issue Date**	*25/10/2026*
Document Code	*FA26SE036_Test Report_v1.0*	**Version**	*1.0*

Record of change

Effective Date	Version	Change Item	*A,D,M	Change description	Reference
25/10/2026	1.0	Whole document	A	Initial test report for Review 2: 75 functional test cases	*Report 3 (SRS), Report 4 (SDD), Design Brief BM01, Report5_Test_Documentation.docx*

TEST CASE LIST

Project Name	*Develop an AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization System (OptiPack AI)*
Project Code	*FA26SE036*
Test Environment Setup Description	*1. Server: back end NestJS 11 on Node.js 22 (CI: ubuntu-latest, GitHub Actions); packer service Python 3.12 / FastAPI (optional)* *2. Database: MongoDB Atlas (replica set, transactions); Redis 7 (Docker)* *3. Web browser: Chrome (latest) for the staff portal (React 19 + Vite) and the AURELLE website (Next.js 15)* *4. External: Lazada test shop, AURELLE Open API simulator (be/scripts/aurelle-mock-server.ts), GHN staging API* *5. Data: seeded with be/scripts/seed-*.ts*

No	Function Name	Sheet Name	Description	Pre-Condition
1	F-01 Authentication and role-based acc	F-01	Acceptance criteria of F-01 (Design Brief	Logged in with a role allowed for the feature
2	F-02 Marketplace connection and order	F-02	Acceptance criteria of F-02 (Design Brief	Logged in with a role allowed for the feature
3	F-03 Order consolidation and groups	F-03	Acceptance criteria of F-03 (Design Brief	Logged in with a role allowed for the feature
4	F-04 Stock reservation and FIFO lots	F-04	Acceptance criteria of F-04 (Design Brief	Logged in with a role allowed for the feature
5	F-05 Picking with missing-item handling	F-05	Acceptance criteria of F-05 (Design Brief	Logged in with a role allowed for the feature
6	F-06 3D packing plan, approval and pac	F-06	Acceptance criteria of F-06 (Design Brief section 6): main flow, data / business rule, exception or permission	Logged in with a role allowed for the feature
7	F-07 Shipping with GHN	F-07	Acceptance criteria of F-07 (Design Brief section 6): main flow, data / business rule, exception or permission	Logged in with a role allowed for the feature
8	F-08 Returns and exchanges	F-08	Acceptance criteria of F-08 (Design Brief section 6): main flow, data / business rule, exception or permission	Logged in with a role allowed for the feature
9	F-09 AURELLE web store and order trac	F-09	Acceptance criteria of F-09 (Design Brief section 6): main flow, data / business rule, exception or permission	Logged in with a role allowed for the feature
10	F-10 Documents and labels	F-10	Acceptance criteria of F-10 (Design Brief section 6): main flow, data / business rule, exception or permission	Logged in with a role allowed for the feature
11	Main workflows (end-to-end)	WF	One end-to-end main scenario per main workflow WF-01 ... WF-06 (Design Brief A1 / A2)	Staging with seeded data; all roles available
12	Order Group state transitions	ST	Every allowed transition of the Order Group (be/src/modules/order-groups/enums/allowed-status-transitions.ts, 33 transitions); invalid jumps are rejected	Order Group in the source status

13	Non-functional requirements	NFR	Each agreed NFR measured with its metric and threshold (Design Brief section 7)	See each case

TEST STATISTICS

Project Name	*Develop an AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization System (OptiPack AI)*	**Creator**	**Nguyễn Phương Mỹ Thuận**	
Project Code	*FA26SE036*	**Reviewer/Approver**	**Thân Thị Ngọc Vân**	
Document Code	*FA26SE036_Test Report_v1.0*	**Issue Date**		*25/10/2026*
Notes	*Review 2 release (tag review-2, commit 1a3101c): 10 mandatory features F-01 … F-10, main workflows WF-01 … WF-06, Order Group state transitions, NFR-1 … NFR-6. Round 1 = automated results of CI run #258; manual cases are pending until the Verification stage.*			

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	F-01 Authentication and role-based access	3	0	0	0	3
2	F-02 Marketplace connection and order sync	3	0	0	0	3
3	F-03 Order consolidation and groups	3	0	0	0	3
4	F-04 Stock reservation and FIFO lots	3	0	0	0	3
5	F-05 Picking with missing-item handling	3	0	0	0	3
6	F-06 3D packing plan, approval and packing	3	0	0	0	3
7	F-07 Shipping with GHN	3	0	0	0	3
8	F-08 Returns and exchanges	3	0	0	0	3
9	F-09 AURELLE web store and order tracking	3	0	0	0	3
10	F-10 Documents and labels	3	0	0	0	3
11	Main workflows (end-to-end)	0	0	6	0	6
12	Order Group state transitions	27	0	6	0	33
13	Non-functional requirements	5	0	1	0	6
	Sub total	62	0	13	0	75

Test coverage	82.67 %
Test successful coverage	82.67 %
Automated in CI	61 of 75 test cases

Feature	F-01 Authentication and role-based access
Test requirement	Acceptance criteria of F-01 (Design Brief section 6): main flow, data / business rule, exception or permission
Number of TCs	3

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-01 Authentication and role-based access										
TC-F01-01	*[Main] Log in with correct e-mail and password (two-factor not enabled)*	*1. Open /login 2. Enter e-mail and password 3. Press Log in*	*Access and refresh tokens are issued; the home screen of the role opens*	*Active staff account, MFA off*	Passed	25/10/2026	GitHub Actions CI	login success without MFA returns a token pair	admin@optipackai.com / valid password	Automated (Jest, CI) – be/src/modules/auth/auth.service.spec.ts:115
TC-F01-02	*[Rule] Wrong password increases the failed-*	*1. Enter a wrong password*	*Each wrong password returns 401 and increments*	*Active staff account*	Passed	25/10/2026	GitHub	wrong password → 401 and counter	Wrong password ×5	Automated (Jest, CI) –
TC-F01-03	*[Permission] A role calls an API outside its*	*1. Log in as Warehouse Staff*	*HTTP 403; the handler is not executed*	*Route protected by @Roles(Admin)*	Passed	25/10/2026	GitHub	route with @Roles(), user with the wrong	Role = WAREHOUSE_STAFF	Automated (Jest, CI) –

Feature	F-02 Marketplace connection and order sync
Test requirement	Acceptance criteria of F-02 (Design Brief section 6): main flow, data / business rule, exception or permission
Number of TCs	3

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-02 Marketplace connection and order sync										
TC-F02-01	*[Main] The scheduler synchronizes a Lazada shop whose sync interval has passed*	*1. Set the sync interval to 30 min and turn AURELLE auto-sync off* *2. Scheduler tick (every minute)* *3. Check which shops are synchronized*	*Only the due and never-polled Lazada shops are synchronized; the not-due shop waits for its interval; AURELLE is not scanned*	*Lazada shops connected*	Passed	25/10/2026	GitHub Actions CI	scheduler syncs only shops that are due and platforms that are enabled	Interval 30 min; shop due (last polled 31 min ago), shop not-due (10 min ago), shop never polled; AURELLE auto-sync off	Automated (Jest, CI) – be/src/modules/orders/order-sync-settings.spec.ts:179
TC-F02-02	*[Rule] The same webhook event delivered twice*	*1. Send a webhook*	*Second delivery is acknowledged (200) without a*	*Event already processed*	Passed	25/10/2026	GitHub	already processed message_id → ack, no	Duplicate message_id	Automated (Jest, CI) – be/src/modules/marketplace-
TC-F02-03	*[Exception] Webhook with an invalid signature is*	*1. Send a webhook with a wrong HMAC signature*	*401 MKT_WEBHOOK_SIGNATURE_INVALID;*	*AURELLE app secret configured*	Passed	25/10/2026	GitHub	wrong signature → 401, no dedupe	Wrong signature	Automated (Jest, CI) – be/src/modules/marketplace-

Feature	F-03 Order consolidation and groups
Test requirement	Acceptance criteria of F-03 (Design Brief section 6): main flow, data / business rule, exception or permission
Number of TCs	3

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-03 Order consolidation and groups										
TC-F03-01	*[Main] A new order joins the open group of the same recipient*	*1. Import order A 2. Import order B with the same phone, address and city*	*Order B is added to the group of order A*	*Group of order A is still open*	Passed	25/10/2026	GitHub Actions CI	sibling in an open group → joined to that group	Same consolidation key	Automated (Jest, CI) – be/src/modules/orders/orders.service.tryConsolidate.spec.ts:93
TC-F03-02	*[Rule] A group that is already picked does not*	*1. Pick group of order A*	*Order B stays standalone (new group)*	*Group status picked or later*	Passed	25/10/2026	GitHub	sibling in a locked group (picked onward)	Same consolidation key	Automated (Jest, CI) –
TC-F03-03	*[Exception] Split a group after an item has been*	*1. Scan one item of the group*	*409 REGROUP_HAS_PICKS; data unchanged*	*Group in picking with a scan*	Passed	25/10/2026	GitHub	already scanned in the current round →	Split request with reason	Automated (Jest, CI) –

Feature	F-04 Stock reservation and FIFO lots
Test requirement	Acceptance criteria of F-04 (Design Brief section 6): main flow, data / business rule, exception or permission
Number of TCs	3

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-04 Stock reservation and FIFO lots										
TC-F04-01	*[Main] A new group reserves available bin stock*	*1. Import an order for a SKU with stock* *2. Check the reservation*	*1 unit reserved, no shortage flag*	*Bin stock 1, nobody reserved*	Passed	25/10/2026	GitHub Actions CI	group 1: bin has 1, nothing reserved → reserves 1, no shortage	Order quantity 1	Automated (Jest, CI) – be/src/modules/order-groups/stock-reservation.service.spec.ts:33
TC-F04-02	*[Rule] Picking consumes the oldest lot first*	*1. Receive lot 1 (2 units) and later lot 2*	*Lot 1 is emptied first, then 1 unit from lot 2*	*Two lots with different dates*	Passed	25/10/2026	GitHub	2 lots, take 3 → old lot (2) first, then 1 from	Lot 1 = 2, lot 2 = 2; pick 3	Automated (Jest, CI) – be/src/modules/warehouse/stock-
TC-F04-03	*[Exception] A second group cannot reserve stock*	*1. Group 1 reserves the last unit*	*Group 2 reserves 0 and is flagged stock shortage*	*Bin stock 1 reserved by group 1*	Passed	25/10/2026	GitHub	group 2: stock already reserved →	Order quantity 1	Automated (Jest, CI) –

Feature	F-05 Picking with missing-item handling
Test requirement	Acceptance criteria of F-05 (Design Brief section 6): main flow, data / business rule, exception or permission
Number of TCs	3

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-05 Picking with missing-item handling										
TC-F05-01	*[Main] Scan the correct SKU of the group*	*1. Open Warehouse Picking* *2. Scan an item of the group*	*Bin stock is decremented and a pick event is written in the same transaction*	*Group in picking, stock in bin*	Passed	25/10/2026	GitHub Actions CI	correct SKU → stock decremented + event in one transaction	SKU of the group	Automated (Jest, CI) – be/src/modules/order-groups/order-groups.service.pickItem.spec.ts:143
TC-F05-02	*[Rule] Scanning more than the ordered quantity is*	*1. Scan a SKU beyond its ordered quantity*	*ORD_GROUP_PICK_EXCEEDS_ORDERED; stock*	*SKU already fully picked*	Passed	25/10/2026	GitHub	scan over ordered quantity →	Extra scan	Automated (Jest, CI) –
TC-F05-03	*[Exception] Confirm picking while items are*	*1. Scan only part of the items*	*ORD_GROUP_PICK_INCOMPLETE with the list of*	*Group in picking*	Passed	25/10/2026	GitHub	confirmPicked with missing items →	Missing SKU	Automated (Jest, CI) –

Feature	F-06 3D packing plan, approval and packing
Test requirement	Acceptance criteria of F-06 (Design Brief section 6): main flow, data / business rule, exception or permission
Number of TCs	3

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-06 3D packing plan, approval and packing										
TC-F06-01	*[Main] Compute packing plans for a picked group*	*1. Group reaches picked* *2. Packing job computes the plan*	*A ready plan for every order, parcels numbered 1..N, group → pending_approval, Packaging Staff notified*	*All SKU profiles confirmed*	Passed	25/10/2026	GitHub Actions CI	ready plan for every order, group → pending_approval	Picked group	Automated (Jest, CI) – be/src/modules/packing/packing-plan.service.spec.ts:275
TC-F06-02	*[Rule] Nothing is placed on top of a fragile item*	*1. Compute a plan for an order with a fragile item*	*No item is placed above the fragile item*	*Fragile SKU in the order*	Passed	25/10/2026	GitHub	fragile item has nothing placed on top	Fragile item + other items	Automated (Jest, CI) –
TC-F06-03	*[Exception] Compute a plan for a group that is not*	*1. Request a plan for a group still in picking*	*PACKING_WRONG_PLAN_STATUS; no plan*	*Group status picking*	Passed	25/10/2026	GitHub	group not fully picked →	Group id	Automated (Jest, CI) – be/src/modules/packing/packing-

Feature	F-07 Shipping with GHN			
Test requirement	Acceptance criteria of F-07 (Design Brief section 6): main flow, data / business rule, exception or permission			
Number of TCs	3			
Testing Round	***Passed***	***Failed***	***Pending***	***N/A***
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-07 Shipping with GHN										
TC-F07-01	*[Main] Create a GHN shipment for a packed group*	*1. Open Shipping & Fulfillment* *2. Choose a quote* *3. Create shipment*	*Shipment awaiting_pickup, each parcel has client_order_code, group → shipped, GHN codes stored*	*Group packed*	Passed	25/10/2026	GitHub Actions CI	shipment starts awaiting pickup, parcels have client_order_code, group → shipped	GHN service, parcels	Automated (Jest, CI) – be/src/modules/shipments/shipments.service.carrier.spec.ts:124
TC-F07-02	*[Rule] Second failed delivery returns the shipment*	*1. Record failed delivery*	*Shipment moves automatically to*	*Shipment out for delivery, manual carrier*	Passed	25/10/2026	GitHub	2nd failure → returning_to_warehouse	Reason: customer unreachable	Automated (Jest, CI) –
TC-F07-03	*[Exception] GHN rejects the address of a parcel*	*1. Create shipment with an invalid address*	*Shipment kept, carrier_booking_error recorded, no*	*Group packed*	Passed	25/10/2026	GitHub	carrier reports a wrong address →	Wrong ward / address	Automated (Jest, CI) –

Feature	F-08 Returns and exchanges
Test requirement	Acceptance criteria of F-08 (Design Brief section 6): main flow, data / business rule, exception or permission
Number of TCs	3

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-08 Returns and exchanges										
TC-F08-01	*[Main] Customer requests a return on AURELLE*	*1. Open Order Detail of a delivered order 2. Press Request return 3. Submit*	*Request created, customer timeline updated, OptiPack notified by webhook*	*Order delivered ≤ 15 days ago*	Passed	25/10/2026	GitHub Actions CI	valid request created and OptiPack notified	Return & refund, 1 item	Automated (Jest, CI) – be/src/modules/storefront/storefront-returns.service.spec.ts:107
TC-F08-02	*[Rule] Return requested more than 15 days after*	*1. Create a return for an order delivered 16 days*	*409 RMA_WINDOW_EXPIRED*	*Delivered order*	Passed	25/10/2026	GitHub	more than 15 days since delivery → 409	Delivered at = today − 16 days	Automated (Jest, CI) –
TC-F08-03	*[Permission] The creator of a return ticket*	*1. User A creates a ticket*	*403 RMA_SELF_APPROVAL*	*Ticket requested*	Passed	25/10/2026	GitHub	creator approves own ticket → 403	Same user	Automated (Jest, CI) –

Feature	F-09 AURELLE web store and order tracking
Test requirement	Acceptance criteria of F-09 (Design Brief section 6): main flow, data / business rule, exception or permission
Number of TCs	3

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-09 AURELLE web store and order tracking										
TC-F09-01	*[Main] A newly placed order shows the 7-step progress bar*	*1. Place an order on AURELLE 2. Open Order Detail*	*7 steps shown, "Placed" is current with the order time*	*Customer logged in*	Passed	25/10/2026	GitHub Actions CI	new order: 7 steps, "Placed" current	New order	Automated (Jest, CI) – be/src/modules/storefront/storefront-order-progress.spec.ts:7
TC-F09-02	*[Rule] An older status event never moves the order*	*1. Send shipped*	*History stored, order status not moved back*	*Order already shipped*	Passed	25/10/2026	GitHub	older event stored but does not move the	Out-of-order events	Automated (Jest, CI) – be/src/modules/storefront/aurelle-
TC-F09-03	*[Exception] Failed delivery is shown with its reason*	*1. GHN reports failed delivery*	*"Delivering to you" step shows an error with the*	*Order out for delivery*	Passed	25/10/2026	GitHub	failed delivery: step shows error with	Failure reason	Automated (Jest, CI) –

Feature	F-10 Documents and labels
Test requirement	Acceptance criteria of F-10 (Design Brief section 6): main flow, data / business rule, exception or permission
Number of TCs	3

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	3	0	0	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ F-10 Documents and labels										
TC-F10-01	*[Main] Print the packing slip of a group*	*1. Open the group* *2. Download packing slip*	*Valid PDF, one page per order*	*Group with orders*	Passed	25/10/2026	GitHub Actions CI	packing slip: valid PDF, one page per order	Group with 2 orders	Automated (Jest, CI) – be/src/modules/documents/pdf/document-renderers.spec.ts:24
TC-F10-02	*[Rule] GHN label: each parcel has its own carrier*	*1. Create a shipment with 2 parcels*	*One page per parcel, each with its own carrier*	*Shipment booked with GHN*	Passed	25/10/2026	GitHub	GHN label: each parcel prints its own	2 parcels	Automated (Jest, CI) –
TC-F10-03	*[Exception] Packing slip of an empty group*	*1. Download packing slip of a group without*	*PDF with one notice page, no error*	*Empty group*	Passed	25/10/2026	GitHub	packing slip: empty group still 1 notice	—	Automated (Jest, CI) –

Feature	Main workflows (end-to-end)			
Test requirement	One end-to-end main scenario per main workflow WF-01 ... WF-06 (Design Brief A1 / A2)			
Number of TCs	6			
Testing Round	***Passed***	***Failed***	***Pending***	***N/A***
Round 1	0	0	6	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ Main workflows (end-to-end)										
TC-WF01	*[End-to-end] WF-01 Order Intake & Consolidation: main scenario end to end*	*1. Create a Lazada order 2. Wait for sync / press Sync (Admin) 3. Open Groups & Express*	*Group created, stock reserved, picker assigned, status picking*	*Lazada shop connected; SKUs mapped and in stock*	Pending		Thuận		Lazada order for 2 SKUs in stock (DS-01)	Manual on the staff portal / AURELLE website with seeded data
TC-WF02	*[End-to-end] WF-02 Picking: main scenario end to*	*1. Log in as Warehouse Staff*	*Group picked; oldest lots consumed*	*Group assigned to the picker*	Pending		Phượng		Group in picking with stock in bins	Manual on the staff portal / AURELLE website with seeded
TC-WF03	*[End-to-end] WF-03 Packing: main scenario end*	*1. Log in as Packaging Staff*	*Group packed; boxes and materials deducted*	*Plan ready*	Pending		Việt		Picked group, SKU profiles confirmed	Manual on the staff portal / AURELLE website with seeded
TC-WF04	*[End-to-end] WF-04 Shipping via GHN: main*	*1. Log in as Shipping Coordinator*	*Shipment and group delivered; AURELLE customer*	*Group packed; GHN staging configured*	Pending		Thuận		Packed group (DS-07)	Manual on the staff portal / AURELLE website with seeded
TC-WF05	*[End-to-end] WF-05 Return / Exchange via*	*1. Customer requests return on AURELLE*	*Ticket closed; refund recorded or replacement*	*Order delivered ≤ 15 days*	Pending		Thi		Delivered AURELLE order within 15	Manual on the staff portal / AURELLE website with seeded
TC-WF06	*[End-to-end] WF-06 Stock Receiving by Lot: main*	*1. Assign SKU to bin*	*Lot created; stock increased; shortage groups re-*	*Warehouse structure created*	Pending		Thi		Warehouse with an empty bin (DS-11)	Manual on the staff portal / AURELLE website with seeded

Feature	Order Group state transitions
Test requirement	Every allowed transition of the Order Group (be/src/modules/order-groups/enums/allowed-status-transitions.ts, 33 transitions); invalid jumps are rejected
Number of TCs	33

Testing Round	*Passed*	*Failed*	*Pending*	*N/A*
Round 1	27	0	6	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ Order Group state transitions										
TC-ST01	*[Transition] Order Group awaiting_packaging → picking*	*1. Group in awaiting_packaging 2. Picker assigned, picking starts*	*Status becomes picking; invalid jumps from awaiting_packaging are rejected (ORD_GROUP state error)*	*Order Group in status awaiting_packaging*	Passed	25/10/2026	GitHub Actions CI	transition asserted (happy path)	Picker assigned, picking starts	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST02	*[Transition] Order Group awaiting_packaging →*	*1. Group in awaiting_packaging*	*Status becomes partial_needs_review; invalid*	*Order Group in status*	Pending		Phượng		Missing item reported before picking	Manual: trigger the action on the portal and check the
TC-ST03	*[Transition] Order Group awaiting_packaging →*	*1. Group in awaiting_packaging*	*Status becomes canceled; invalid jumps from*	*Order Group in status*	Passed	25/10/2026	GitHub	transition asserted (auto-cancel)	All orders cancelled	Automated (Jest, CI) –
TC-ST04	*[Transition] Order Group picking → picked*	*1. Group in picking*	*Status becomes picked; invalid jumps from picking*	*Order Group in status picking*	Passed	25/10/2026	GitHub	transition asserted (happy path)	Complete picking	Automated (Jest, CI) –
TC-ST05	*[Transition] Order Group picking →*	*1. Group in picking*	*Status becomes partial_needs_review; invalid*	*Order Group in status picking*	Pending		Phượng		Report missing item	Manual: trigger the action on the portal and check the
TC-ST06	*[Transition] Order Group picking → canceled*	*1. Group in picking*	*Status becomes canceled; invalid jumps from*	*Order Group in status picking*	Passed	25/10/2026	GitHub	transition asserted (auto-cancel)	All orders cancelled	Automated (Jest, CI) –
TC-ST07	*[Transition] Order Group picked →*	*1. Group in picked*	*Status becomes pending_approval; invalid jumps*	*Order Group in status picked*	Passed	25/10/2026	GitHub	transition asserted (happy path)	Packing plan computed	Automated (Jest, CI) –
TC-ST08	*[Transition] Order Group picked → picking*	*1. Group in picked*	*Status becomes picking; invalid jumps from picked*	*Order Group in status picked*	Pending		Việt		Back to picking from packing	Manual: trigger the action on the portal and check the
TC-ST09	*[Transition] Order Group picked → canceled*	*1. Group in picked 2. All orders cancelled*	*Status becomes canceled; invalid jumps from picked are rejected (ORD_GROUP state error)*	*Order Group in status picked*	Passed	25/10/2026	GitHub Actions CI	transition asserted (auto-cancel)	All orders cancelled	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST10	*[Transition] Order Group partial_needs_review → picked*	*1. Group in partial_needs_review 2. Ship available items*	*Status becomes picked; invalid jumps from partial_needs_review are rejected (ORD_GROUP state error)*	*Order Group in status partial_needs_review*	Passed	25/10/2026	GitHub Actions CI	transition asserted (partial → picked)	Ship available items	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST11	*[Transition] Order Group partial_needs_review → awaiting_packaging*	*1. Group in partial_needs_review 2. Re-queue until stock arrives*	*Status becomes awaiting_packaging; invalid jumps from partial_needs_review are rejected (ORD_GROUP state error)*	*Order Group in status partial_needs_review*	Passed	25/10/2026	GitHub Actions CI	transition asserted (partial → awaiting)	Re-queue until stock arrives	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST12	*[Transition] Order Group partial_needs_review → picking*	*1. Group in partial_needs_review 2. Re-pick (new pick round)*	*Status becomes picking; invalid jumps from partial_needs_review are rejected (ORD_GROUP state error)*	*Order Group in status partial_needs_review*	Pending		Phượng		Re-pick (new pick round)	Manual: trigger the action on the portal and check the group status
TC-ST13	*[Transition] Order Group partial_needs_review → canceled*	*1. Group in partial_needs_review 2. All orders cancelled*	*Status becomes canceled; invalid jumps from partial_needs_review are rejected (ORD_GROUP state error)*	*Order Group in status partial_needs_review*	Passed	25/10/2026	GitHub Actions CI	transition asserted (auto-cancel)	All orders cancelled	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST14	*[Transition] Order Group pending_approval → approved_for_packing*	*1. Group in pending_approval 2. Approve plan*	*Status becomes approved_for_packing; invalid jumps from pending_approval are rejected (ORD_GROUP state error)*	*Order Group in status pending_approval*	Passed	25/10/2026	GitHub Actions CI	transition asserted (happy path)	Approve plan	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST15	*[Transition] Order Group pending_approval → picked*	*1. Group in pending_approval 2. Reject plan / recompute*	*Status becomes picked; invalid jumps from pending_approval are rejected (ORD_GROUP state error)*	*Order Group in status pending_approval*	Passed	25/10/2026	GitHub Actions CI	transition asserted (reject plan)	Reject plan / recompute	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST16	*[Transition] Order Group pending_approval → canceled*	*1. Group in pending_approval 2. All orders cancelled*	*Status becomes canceled; invalid jumps from pending_approval are rejected (ORD_GROUP state error)*	*Order Group in status pending_approval*	Passed	25/10/2026	GitHub Actions CI	transition asserted (auto-cancel)	All orders cancelled	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST17	*[Transition] Order Group approved_for_packing → packed*	*1. Group in approved_for_packing 2. Last parcel sealed*	*Status becomes packed; invalid jumps from approved_for_packing are rejected (ORD_GROUP state error)*	*Order Group in status approved_for_packing*	Passed	25/10/2026	GitHub Actions CI	transition asserted (happy path)	Last parcel sealed	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST18	*[Transition] Order Group approved_for_packing → pending_approval*	*1. Group in approved_for_packing 2. Undo approval*	*Status becomes pending_approval; invalid jumps from approved_for_packing are rejected (ORD_GROUP state error)*	*Order Group in status approved_for_packing*	Passed	25/10/2026	GitHub Actions CI	transition asserted (undo approve)	Undo approval	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST19	*[Transition] Order Group approved_for_packing → picked*	*1. Group in approved_for_packing 2. Recompute after approval*	*Status becomes picked; invalid jumps from approved_for_packing are rejected (ORD_GROUP state error)*	*Order Group in status approved_for_packing*	Pending		Việt		Recompute after approval	Manual: trigger the action on the portal and check the group status
TC-ST20	*[Transition] Order Group approved_for_packing → canceled*	*1. Group in approved_for_packing 2. All orders cancelled*	*Status becomes canceled; invalid jumps from approved_for_packing are rejected (ORD_GROUP state error)*	*Order Group in status approved_for_packing*	Passed	25/10/2026	GitHub Actions CI	transition asserted (auto-cancel)	All orders cancelled	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST21	*[Transition] Order Group approved_for_packing → picking*	*1. Group in approved_for_packing 2. Damaged / missing item while packing, back to picking*	*Status becomes picking; invalid jumps from approved_for_packing are rejected (ORD_GROUP state error)*	*Order Group in status approved_for_packing*	Passed	25/10/2026	GitHub Actions CI	transition asserted (back to picking)	Damaged / missing item while packing, back to picking	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST22	*[Transition] Order Group packed → shipped*	*1. Group in packed 2. Shipment created*	*Status becomes shipped; invalid jumps from packed are rejected (ORD_GROUP state error)*	*Order Group in status packed*	Passed	25/10/2026	GitHub Actions CI	transition asserted (happy path)	Shipment created	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST23	*[Transition] Order Group packed → canceled*	*1. Group in packed 2. Orders cancelled after packing (unpack parcels)*	*Status becomes canceled; invalid jumps from packed are rejected (ORD_GROUP state error)*	*Order Group in status packed*	Passed	25/10/2026	GitHub Actions CI	transition asserted (packed → canceled)	Orders cancelled after packing (unpack parcels)	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST24	*[Transition] Order Group packed → approved_for_packing*	*1. Group in packed 2. Reopen / repack a parcel*	*Status becomes approved_for_packing; invalid jumps from packed are rejected (ORD_GROUP state error)*	*Order Group in status packed*	Pending		Thuận		Reopen / repack a parcel	Manual: trigger the action on the portal and check the group status
TC-ST25	*[Transition] Order Group shipped → delivered*	*1. Group in shipped 2. Carrier delivered*	*Status becomes delivered; invalid jumps from shipped are rejected (ORD_GROUP state error)*	*Order Group in status shipped*	Passed	25/10/2026	GitHub Actions CI	transition asserted (happy path)	Carrier delivered	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST26	*[Transition] Order Group shipped → returned*	*1. Group in shipped 2. Returned to warehouse after failed delivery*	*Status becomes returned; invalid jumps from shipped are rejected (ORD_GROUP state error)*	*Order Group in status shipped*	Passed	25/10/2026	GitHub Actions CI	transition asserted (shipped → returned)	Returned to warehouse after failed delivery	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST27	*[Transition] Order Group shipped → packed*	*1. Group in shipped 2. GHN order cancelled before pickup*	*Status becomes packed; invalid jumps from shipped are rejected (ORD_GROUP state error)*	*Order Group in status shipped*	Passed	25/10/2026	GitHub Actions CI	transition asserted (shipped → packed)	GHN order cancelled before pickup	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST28	*[Transition] Order Group delivered → returning*	*1. Group in delivered 2. Full return ticket created*	*Status becomes returning; invalid jumps from delivered are rejected (ORD_GROUP state error)*	*Order Group in status delivered*	Passed	25/10/2026	GitHub Actions CI	transition asserted (returning)	Full return ticket created	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST29	*[Transition] Order Group delivered → returned*	*1. Group in delivered 2. Return received after delivery*	*Status becomes returned; invalid jumps from delivered are rejected (ORD_GROUP state error)*	*Order Group in status delivered*	Passed	25/10/2026	GitHub Actions CI	transition asserted (delivered → returned)	Return received after delivery	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST30	*[Transition] Order Group returning → returned*	*1. Group in returning 2. Returned goods received*	*Status becomes returned; invalid jumps from returning are rejected (ORD_GROUP state error)*	*Order Group in status returning*	Passed	25/10/2026	GitHub Actions CI	transition asserted (next statuses of returning)	Returned goods received	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST31	*[Transition] Order Group returning → delivered*	*1. Group in returning 2. Return ticket rejected / cancelled*	*Status becomes delivered; invalid jumps from returning are rejected (ORD_GROUP state error)*	*Order Group in status returning*	Passed	25/10/2026	GitHub Actions CI	transition asserted (next statuses of returning)	Return ticket rejected / cancelled	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST32	*[Transition] Order Group returned → canceled*	*1. Group in returned 2. Inspection done and stock handled*	*Status becomes canceled; invalid jumps from returned are rejected (ORD_GROUP state error)*	*Order Group in status returned*	Passed	25/10/2026	GitHub Actions CI	transition asserted (next statuses of returned)	Inspection done and stock handled	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts
TC-ST33	*[Transition] Order Group returned → delivered*	*1. Group in returned 2. Return ticket cancelled*	*Status becomes delivered; invalid jumps from returned are rejected (ORD_GROUP state error)*	*Order Group in status returned*	Passed	25/10/2026	GitHub Actions CI	transition asserted (next statuses of returned)	Return ticket cancelled	Automated (Jest, CI) – be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts

Feature	Non-functional requirements			
Test requirement	Each agreed NFR measured with its metric and threshold (Design Brief section 7)			
Number of TCs	6			
Testing Round	***Passed***	***Failed***	***Pending***	***N/A***
Round 1	5	0	1	0
Round 2	0	0	0	0
Round 3	0	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Note	Test data	Run method / evidence
▶ **Non-functional requirements**										
TC-NFR1	*[NFR] New Lazada order visible in OptiPack*	*1. Create a Lazada test order and note the time* *2. Read the sync job log time*	*≤ sync interval + 1 min (≤ 6 min)*	*Shop sync interval = 5 min*	Pending		Thuận			Manual measurement on staging (Lazada test shop)
TC-NFR2	*[NFR] Packing plan computation time*	*1. Run be/scripts/solver-benchmark.ts on the 300-*	*BRKGA p95 ≤ 1 s per order*	*300-order benchmark data*	Passed	04/10/2026	Thi	be/scripts/benchmark-results/solver-		Automated script (not in CI) – be/scripts/solver-
TC-NFR3	*[NFR] Lockout and role checks*	*1. Wrong password ×5*	*Account locked 15 min; 403 for a wrong role*	*Active account*	Passed	25/10/2026	GitHub	locked account → 403; wrong role → 403		Automated (Jest, CI) –
TC-NFR4	*[NFR] Status write-back retry*	*1. AURELLE returns an error*	*Retry with back-off, no row lost; failed after 8*	*Outbox row queued*	Passed	25/10/2026	GitHub	error from marketplace → attempts+1,		Automated (Jest, CI) – be/src/modules/marketplace-
TC-NFR5	*[NFR] Repeated scan is not counted twice*	*1. Send the same pick scan twice with the same*	*Second scan returns the first result; stock*	*Group in picking*	Passed	25/10/2026	GitHub	client_event_id already processed →		Automated (Jest, CI) –
TC-NFR6	*[NFR] Packing plan without the CP-SAT service*	*1. Stop the packer service*	*Plan still produced (BRKGA); cp_sat = unavailable*	*PACKER_URL not reachable*	Passed	25/10/2026	GitHub	no CP-SAT: plan still produced, cp_sat =		Automated (Jest, CI) – be/src/modules/packing/packing-